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 / [Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

Started on Wednesday, 23 October 2024, 12:15 PM

State Finished

Completed on Wednesday, 23 October 2024, 12:36 PM

Time taken 21 mins 39 secs

Marks 3.00/3.00

Grade 3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

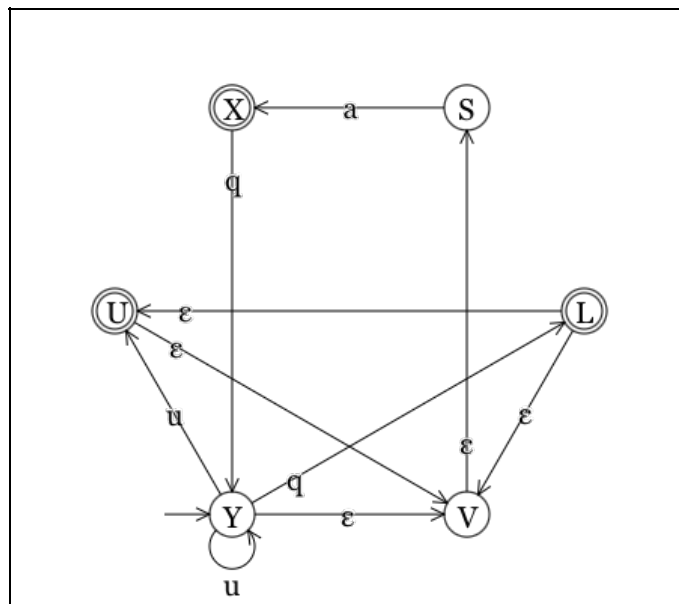
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

Translate the following DFA to a right-regular grammar.



Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals.

Your answers must be syntactically correct for marking.

Your last answer was interpreted as:

Terminals = { "a", "q", "u" }

$\{a, q, u\}$

Nonterminals = { "X", "S", "L", "V", "Y", "U" }

Your last answer was interpreted as:

$\{X, S, L, V, Y, U\}$

Start symbol "Y"

Your last answer was interpreted as:

Y

Rules for L = { ["L", ["V"]], ["L", ["U"]], ["L", []] }

Your last answer was interpreted as

 $\{[L, [V]], [L, [U]], [L,$

Rules for V = { ["V", ["S"]] }

Your last answer was interpreted as:

$$\{[V, [S]]\}$$

Rules for Y = { ["Y", ["u", "Y"]], ["Y", ["V"]], ["Y", ["q", "L"]], ["Y"

Your last answer was interpreted as:

$$\{[Y, [u, Y]], [Y, [V]], [Y, [q, L]]\}$$

Rules for U = { ["U", ["V"]], ["U", []] }

Your last answer was interpreted as:

 $\{[U, [V]], [U, []]\}$

Rules for X = { ["X", ["q", "Y"]], ["X", []] }

Your last answer was interpreted as:

$$\{ [X, [q, Y]] , [X, []]$$

Rules for S = { ["S", ["a", "X"]] }

Your last answer was interpreted as

$$\{[S, [a, X]]\}$$

Question 2

Correct

Mark 1.00 out of 1.00

Which temporal logic statement matches the English sentence "Some love leads to a continuous marriage."?

$$F(\text{love} \Rightarrow G \text{ marriage})$$

Question 3

Correct

Mark 1.00 out of 1.00

Consider the following relation R over integers.

$$xRy \text{ iff } 3 + x \% 6 = 3 + y \% 6$$

Check the properties of the relation R.

The relation is reflexive:	True
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True

The relation is symmetric:	True
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True

The grammar is antisymmetric: False

False

The grammar is transitive:	True
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True

Question **4**

Correct

Mark 0.00 out
of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 151

Signing code:

Your last answer was interpreted as follows:

15368

◀ [Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)